The smaller group censuses in the regular-polytope algorithm paper

Human Compression project: L071 recovery

Working application; complete validation and distinct review pending

Leemans and Mixer, *Algorithms for classifying regular polytopes with a fixed automorphism group* (2012), give twelve numerical comparisons on physical page 4 and verify the earlier sporadic classifications on pages 9–10. We recover these finite classifications from the acquired theorem on enumeration of marked groups and its implementation by classes in centralizers. The alternating groups $A_{11}, \ldots, A_{14}$, the complete Co_3 tables and the correction to the compatible-action criterion are treated in separate components of this same recovery.

An ordered class means a class of tuples under the full automorphism group of the generated group. Reversal is not part of that equivalence. Write c for the number of ordered classes, s for the number fixed by reversal, and $d = (c + s)/2$ for the number after identifying duals. The standard correspondence with regular polytopes is retained throughout.

Theorem 1. *For the ten groups in Table 1, the numbers of regular polytopes are the displayed c, s, d. In the twelve-group comparison of the source, the other two entries are as follows. The group* $\mathrm{Sz}(8)$ *has seven classes up to duality, all of rank three, and none is self-dual. The group $A_5 \times C_2$ has eleven classes up to duality: eight of rank three with connected diagrams and three of rank four with disconnected diagrams. Thus its printed count eight requires the connected-diagram restriction.*

Group	c	s	d
A_5	3	1	2
$\mathrm{P\Gamma L}_2(9)$	21	3	12
S_7	81	7	44
$\mathrm{PSL}_2(25)$	30	4	17
$\mathrm{P\Sigma U}_3(3)$	60	2	31
$\mathrm{PGL}_2(27)$	190	6	98
M_{12}	67	7	37
J_1	300	0	150
A_9	84	10	47
S_9	352	12	182

Table 1: Complete classes for the benchmark groups. The last column is the equivalence convention of the source's first comparison table.

Theorem 2. *For the eleven sporadic simple groups of order at most* $|\mathrm{He}|$*, the ordered class counts by rank are those in Table 2. All other ranks have zero classes. The same statement, with the separately displayed counts, holds for the Tits group ${}^2F_4(2)'$. The complete lists supplied by Hartley and Hulpke are in bijection with these independently enumerated ordered classes, through explicit isomorphisms and conjugations of their actual generators.*

Group	Rank 3	Rank 4	Rank 5	Total
M_{11}	0	0	0	0
M_{12}	40	27	0	67
J_1	296	4	0	300
M_{22}	0	0	0	0
J_2	261	31	0	292
M_{23}	0	0	0	0
HS	465	111	4	580
M_{24}	946	310	4	1260
McL	0	0	0	0
He	2292	145	0	2437
J_3	584	2	0	586
${}^2F_4(2)'$	468	57	0	525

Table 2: Ordered classes. The Tits group is a separate additional classification, not one of the twenty-six sporadic simple groups.

Common proof of the finite census statements. For each concrete permutation model, construct a normalizing overgroup A of G. The retained executions check the expected order of G, normality in A, and

$$|A : C_A(G)| = |\operatorname{Aut}(G)|.$$

Thus the conjugation action really supplies every automorphism, as required by the acquired theorem; containment of two unrelated group embeddings is not being used. The identifications of the concrete library models with the named groups remain explicit standard inputs. For M_{12} we use degree 144 to implement the outer automorphism. For McL the group is the derived subgroup of the chosen automorphism overgroup, rather than a separately stored subgroup in unrelated coordinates. The J_3 overgroup is supplied by the retained Atlas generators in degree 6156.

The acquired initial-segment theorem enumerates one representative of each orbit of possible next marks under the pointwise stabilizer of the current tuple. Its intersection test is applied only to extensions of prefixes already known to have the intersection property. The complete theorem for classes in centralizers proves that the concrete double-coset construction supplies exactly these candidate orbits. Its necessary-generation test uses the full normal closure of the eligible involution classes, as that proof requires.

Every new independent mark at least doubles the generated subgroup. Consequently a string C-marking of G has rank at most $\lfloor \log_2 |G| \rfloor$. The search computes this bound from the group order and treats every rank through that bound. It reads no source rank cutoff or expected count. Each of the twenty model runs terminates normally with the complete-exhaustion record and unchanged inputs. Their union contains 6868 ordered tuples. The acquired completeness theorem therefore applies to the actual complete outputs, including the four groups for which every output list is empty.

The retained validation checks every actual tuple: each mark is an involution in G, the marks generate G, the string commutations hold, and the intersection equality holds for every pair of subsets of the marking. It computes the orders of every adjacent product, giving the complete ordered type distributions. This validation supplements the enumeration theorem; it does not infer exhaustion from the number of surviving tuples. Reversal is tested in the full automorphism action, which yields s and hence d.

The two reused benchmark rows require no further search. The acquired complete H_3 census applies because $H_3 \cong A_5 \times C_2$ and gives exactly the eleven stated classes. The acquired Suzuki

counting theorem gives

$$\frac{1}{2 \cdot 3} \sum_{e \mid 3} \mu(3/e)(2^e - 2)(2^e - 1) = 7.$$

Its complete rank theorem excludes every other rank and its duality argument excludes self-dual classes. Both entire acquired proofs and their qualifications remain dependencies of this application.

Finally the source comparison uses all twelve original auxiliary lists, containing 6047 ordered markings. An isomorphism maps each source group to our model, and a recorded automorphism then maps each source tuple to one specified enumerated tuple. The terminal tuples are distinct, and the two lists have equal cardinality for every group. The acquired comparison theorem therefore gives a bijection of marked classes. This proves the assertion about the original lists themselves, including their ordered types; equality of the three rank totals alone would not prove that assertion. □

Source conventions and attribution. The eleven ordinary sporadic groups are the literal scope of the source's pages 9–10. Its broader description on page 2 refers to the complete Hartley–Hulpke classifications, whose auxiliary archive also contains the Tits group. We include the latter explicitly under that broader reading. The twelve benchmark entries contain two of the ordinary sporadic groups, M_{12} and J_1; they are computed and counted only once in the twenty-model execution. The two reused groups complete the twenty-two distinct groups in this component.

The lists of Hartley and Hulpke and the numerical results of Leemans and Mixer retain their original attribution. Our contribution is the application of the common enumeration theorem, complete independent executions, and explicit correspondence with the original generators. Historical CPU times and memory use are measurements of the authors' implementations and hardware; this proof does not reproduce them or claim a performance comparison. The source's earlier A_{10} work and its open-question remark are outside the new finite classifications proved here. No open-conjecture answer or complete Lean formalization is claimed by these computations.